SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: OBJECT ORIENTED ANALYSIS AND DESIGN
COURSE CODE:CSA1104

COURSE OUTCOME COVERED

CO2: Design UML diagrams to represent system requirements and architecture. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 6 **Total Marks:60**

Annexure A

Application Based Questions

Code of Conduct:

I __Dharshini/192521408_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:
Dharshini

Annexure A

Application Based Questions

1. Use Case Diagram Analysis

A **Smart Pharmacy Management System** allows customers to search medicines, upload prescriptions, place medicine orders, and make online payments. Pharmacists verify prescriptions and dispense medicines, while administrators manage inventory and user accounts. The current system design does not clearly define actor responsibilities and interactions.
Analyze the system requirements, identify all actors and missing use cases, and design an optimized **Use Case Diagram** showing appropriate relationships (**include, extend, and generalization**) to improve system functionality.
(10 Marks)

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2. Class Diagram Design

A **Smart Library Management System** consists of entities such as Book, Member, Librarian, Reservation, Fine, and BorrowRecord. The existing system does not clearly specify class attributes, operations, or relationships among these entities.
Analyze the system, identify relevant classes with appropriate attributes and methods, define relationships (**association, aggregation, composition, and inheritance**), specify multiplicity, and construct a well-structured **Class Diagram**.
(10 Marks)

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3. Sequence Diagram Evaluation

An **Online Examination System** enables students to log in, select an examination, answer questions, submit responses, and receive results. The communication among the student, examination module, evaluation module, and database is incomplete.
Analyze the interaction process, identify all participating objects and missing message exchanges, and design a detailed **Sequence Diagram** representing the complete interaction flow with activation bars and lifelines.
(10 Marks)

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4. Activity Diagram Construction

A **Loan Approval System** allows customers to submit loan applications, upload supporting documents, complete eligibility verification, receive loan approval or rejection, and complete loan disbursement. The workflow lacks proper decision points and parallel activities.
Analyze the business process, identify all workflow activities, include decision nodes for approval and rejection, represent parallel processing where applicable, and construct an optimized **Activity Diagram**.
(10 Marks)

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5. Component Diagram Design

A **Smart E-Commerce Platform** consists of modules such as User Management, Product Catalog, Shopping Cart, Order Management, Payment Gateway, Inventory Management, and Notification Service. The existing architecture does not clearly define interfaces and dependencies among modules.
Analyze the software architecture, identify major components and their provided/required interfaces, define dependencies among components, and design a comprehensive **Component Diagram** representing the overall system architecture.
(10 Marks)

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6. Deployment Diagram Design

A **Smart Agriculture Monitoring System** uses IoT soil sensors, weather stations, irrigation controllers, cloud servers, mobile applications, and farmer dashboards to monitor crop health and automate irrigation. The deployment architecture lacks proper representation of hardware nodes and communication links.
Analyze the deployment environment, identify hardware nodes and deployed software artifacts, specify communication protocols between nodes, and construct a detailed **Deployment Diagram** representing the complete physical system architecture.
(10 Marks)

Answer:

1. Use Case Diagram Analysis – Smart Pharmacy Management System

Actors

- Customer: Searches medicines, uploads prescriptions, places orders, and makes payments.
- Pharmacist: Verifies prescriptions and dispenses medicines.
- Administrator: Manages medicine inventory and user accounts.
- Payment Gateway: Processes online payments.

Use cases

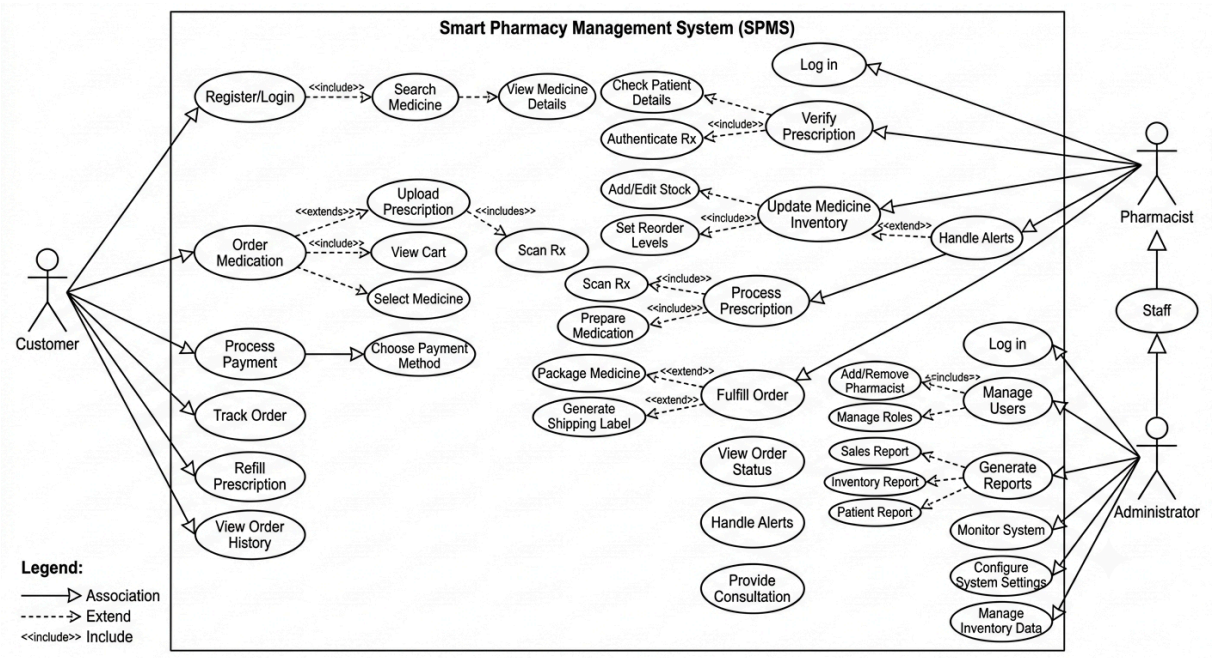
- Register/Login
- Search Medicines
- View Medicine Details

- Upload Prescription
- Place Medicine Order
- Verify Prescription
- Check Medicine Availability
- Make Online Payment
- Dispense Medicines
- Manage Inventory
- Manage User Accounts
- Send Order Confirmation

Relationships

- Include: Place Medicine Order includes checking medicine availability.
- Include: Place Medicine Order includes online payment processing.
- Extend: Prescription verification extends the order process when a medicine requires a prescription.
- Generalization: Customer and Pharmacist can be specialized types of a general User actor, sharing common login functionality.

Use Case Diagram



The diagram separates customer, pharmacist, administrator, and payment gateway responsibilities. Including availability checking helps prevent orders for unavailable medicines, while conditional prescription verification supports safe medicine dispensing.

2. Class Diagram Design – Smart Library Management System

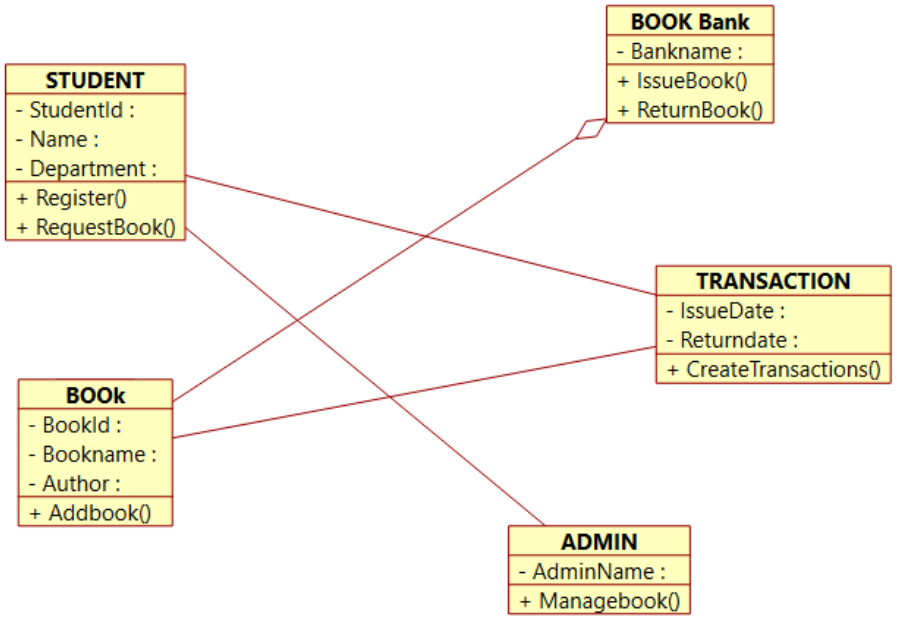
Classes, attributes, and methods

Class	Attributes	Methods
Book	bookId, title, author, ISBN, availability	checkAvailability(), updateAvailability()
Member	memberId, name, email, phone	searchBook(), reserveBook(), borrowBook(), returnBook()
Librarian	librarianId, name, email	addBook(), removeBook(), issueBook(), collectFine()
Reservation	reservationId, reservationDate, status	createReservation(), cancelReservation()
Fine	fineId, amount, paymentStatus	calculateFine(), payFine()
BorrowRecord	recordId, issueDate, dueDate, returnDate	createRecord(), closeRecord(), calculateOverdue()

Relationships and multiplicity

- Association: A Member borrows books through BorrowRecord.
- Association: A Librarian manages borrowing and returning activities.
- Aggregation: A Library contains Books, but books can exist independently of a particular library collection.
- Composition: A BorrowRecord contains its associated Fine record in this simplified design; the fine record is managed as part of that borrowing transaction.
- Inheritance: Member and Librarian inherit common attributes such as ID, name, and contact details from a general Person class.
- One Member can have 0..* BorrowRecords.
- One Book can appear in 0..* BorrowRecords over time.
- One BorrowRecord can have 0..1 Fine.

Class Diagram



The class diagram organizes library operations into reusable classes. Inheritance reduces duplicated person details, associations represent borrowing and reservations, and multiplicity defines how many records can be connected to each object.

3. Sequence Diagram Evaluation – Online Examination System

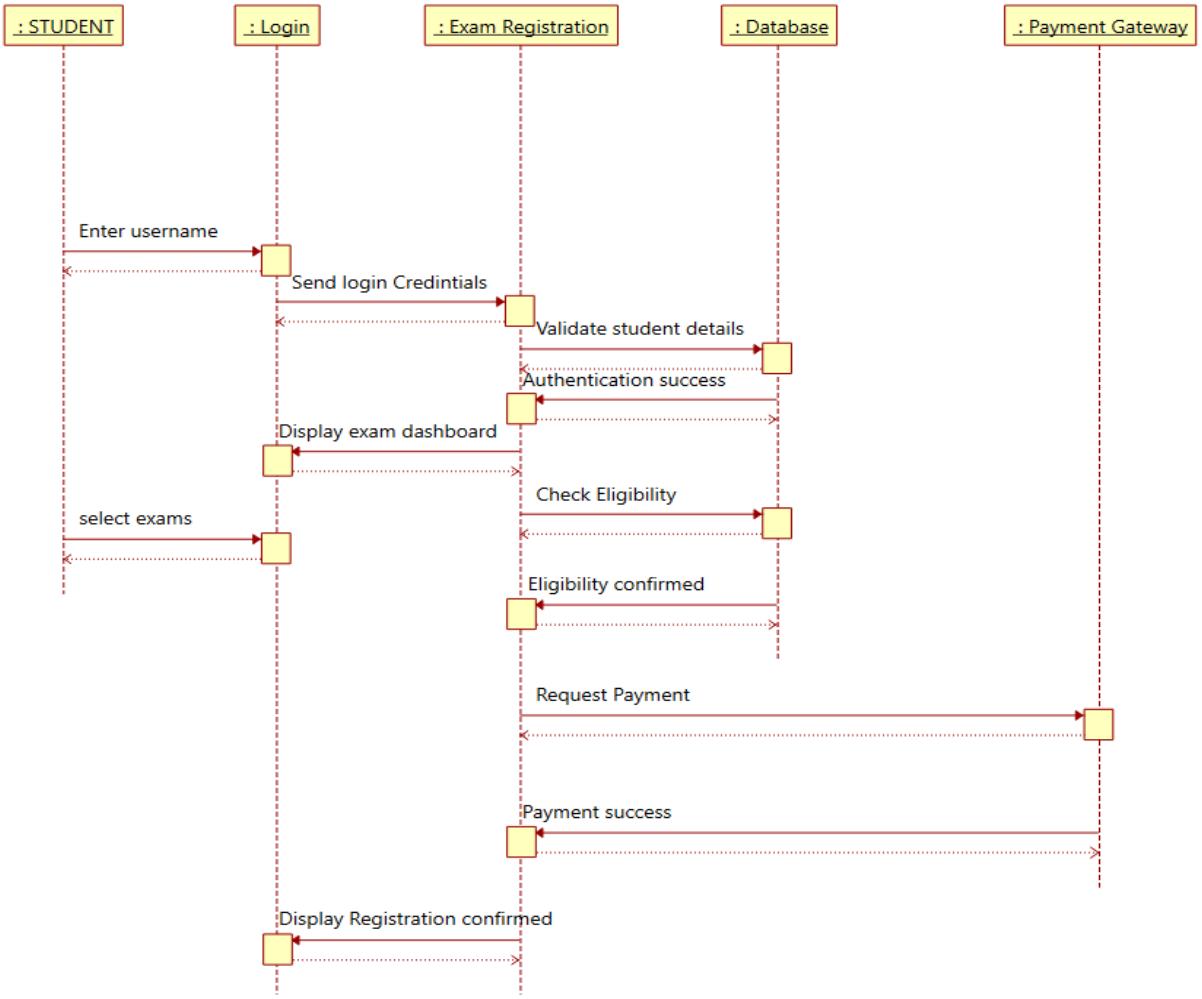
Participating objects

- Student: Logs in, selects an examination, answers questions, and submits responses.
- Login Module: Authenticates the student.
- Examination Module: Displays examination details and questions.
- Evaluation Module: Evaluates submitted answers and calculates marks.
- Database: Stores student information, questions, responses, and results.

Message flow

1. The student enters login credentials.
2. The Login Module validates the credentials using the database.
3. The student selects an examination.
4. The Examination Module retrieves questions from the database and displays them.
5. The student answers questions and submits responses.
6. The Examination Module saves the responses.
7. The Evaluation Module retrieves the responses and evaluates them.
8. The results are stored in the database and displayed to the student.

Sequence Diagram



Explanation: The sequence diagram shows the complete examination process in chronological order. Lifelines represent the participating objects, activation bars indicate active processing, and the alternative fragment handles successful and failed login attempts.

4. Activity Diagram Construction – Loan Approval System

Workflow activities

- Customer registers or logs in.
- Customer fills in the loan application.
- Customer uploads supporting documents.
- The system validates the application.
- Eligibility verification is performed.
- Documents and financial details are reviewed.
- The loan is approved or rejected.
- The customer receives the decision.
- Approved loans proceed to disbursement.

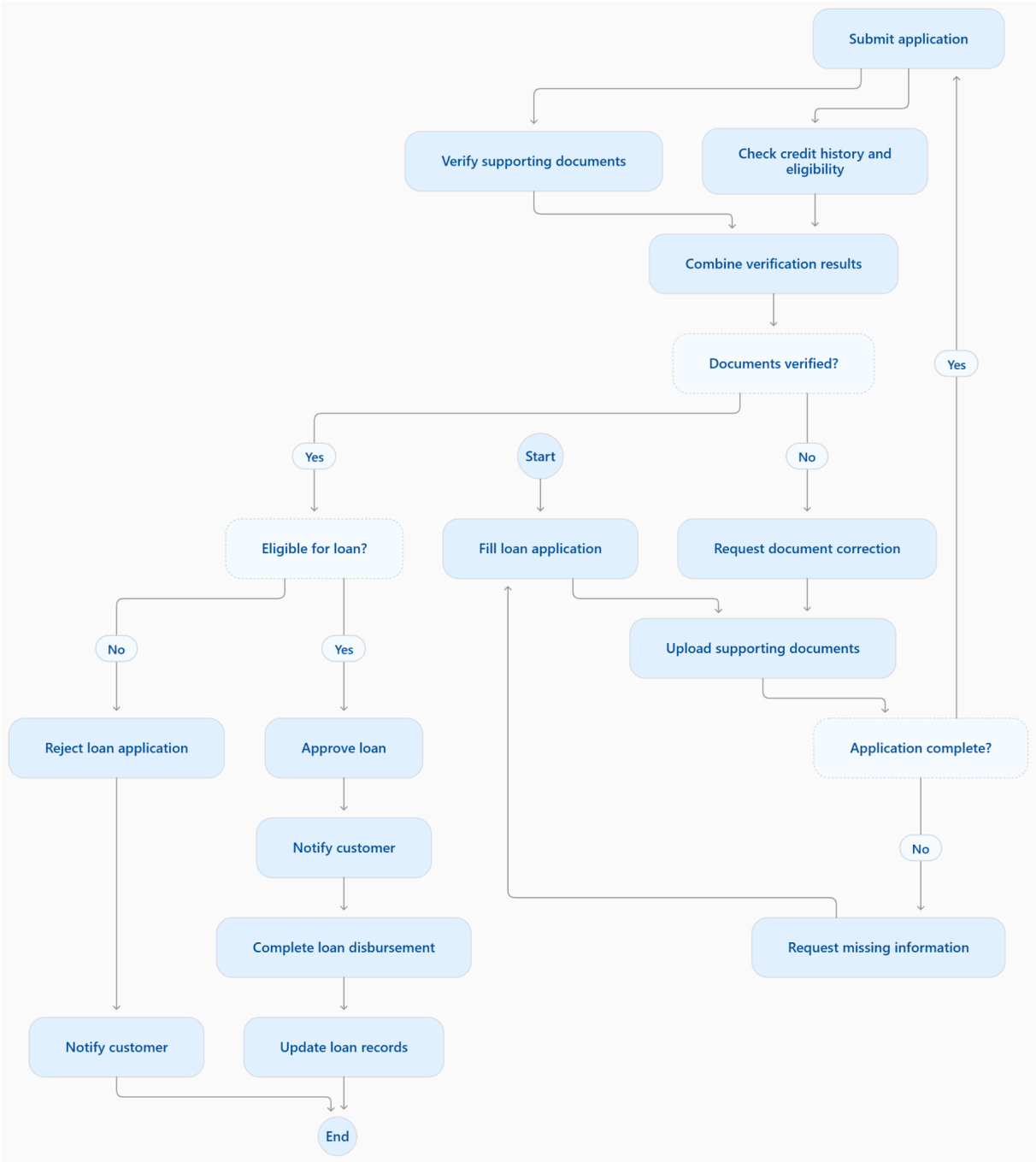
Decision nodes

- Application complete? — If no, the customer must provide missing information.
- Eligible for loan? — If no, the application is rejected.
- Documents verified? — If no, additional documents or corrections may be requested.
- Loan approved? — Determines whether the loan proceeds to disbursement.

Parallel processing

After submission, the system can check credit history and verify supporting documents in parallel. The results are combined before the final approval decision. This reduces processing time.

Activity Diagram



The activity diagram includes decisions, rejection paths, and parallel verification activities. Processing credit history and documents simultaneously reduces delays, while the decision nodes ensure that only eligible applicants with verified documents proceed to loan approval and disbursement.

5. Component Diagram Design – Smart E-Commerce Platform

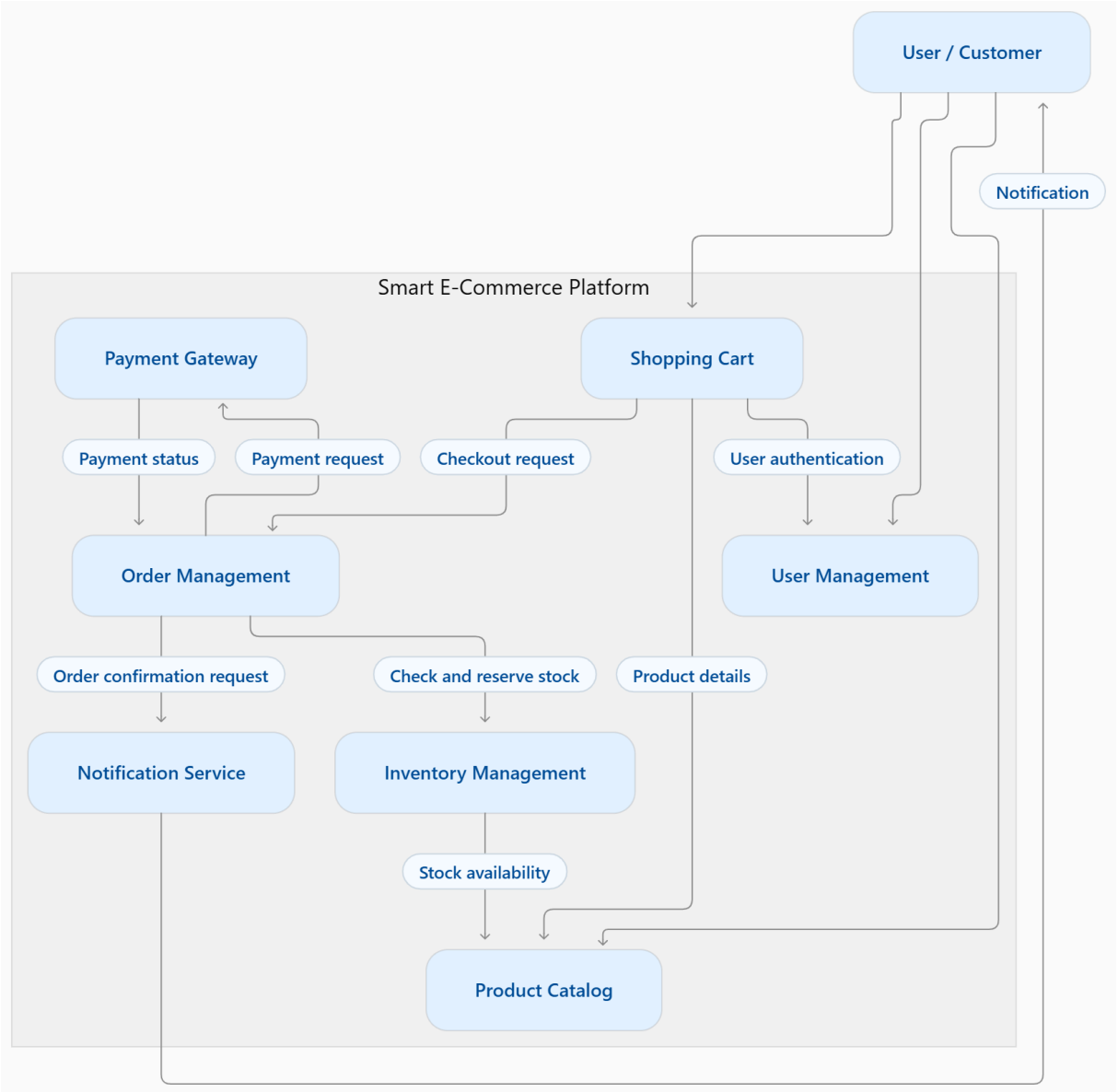
Major components

- User Management: Handles registration, login, and authentication.
- Product Catalog: Displays products, prices, and product details.
- Shopping Cart: Adds, removes, and manages selected products.
- Order Management: Creates orders and tracks order status.
- Payment Gateway: Processes online payments.
- Inventory Management: Checks and updates available stock.
- Notification Service: Sends order confirmations and payment notifications.

Interfaces and dependencies

Component	Required interface or dependency
User Management	Authentication interface
Product Catalog	Product information interface
Shopping Cart	Product Catalog and User Management
Order Management	Shopping Cart and Inventory Management
Payment Gateway	Payment processing interface
Inventory Management	Stock checking and updating interface
Notification Service	Email, SMS, or push notification interface

Component Diagram



The component diagram shows how independent software modules communicate through defined interfaces. The Shopping Cart depends on the Product Catalog to obtain product information, while Order Management coordinates stock reservation, payment, and order confirmation. Separating components makes the platform easier to maintain, test, and extend.

6. Deployment Diagram Design – Smart Agriculture Monitoring System

Hardware nodes

- IoT Soil Sensors: Measure soil moisture.
- Weather Station: Measures temperature, humidity, and other supported weather conditions.
- IoT Gateway: Collects readings and forwards data to the cloud.
- Cloud Server: Stores and processes sensor data and runs irrigation decision logic.
- Irrigation Controller: Receives commands and controls irrigation equipment.
- Mobile Device: Runs the farmer's monitoring application.
- Farmer Dashboard: Provides a larger interface for monitoring crop health; it may run on a computer or browser.

Deployed software artifacts

Hardware node	Software artifacts
Soil Sensors	Sensor firmware and data collection drivers
Weather Station	Weather measurement firmware
IoT Gateway	Gateway firmware and communication service
Cloud Server	Data collection API, database, crop monitoring service, irrigation decision service
Irrigation Controller	Irrigation control firmware

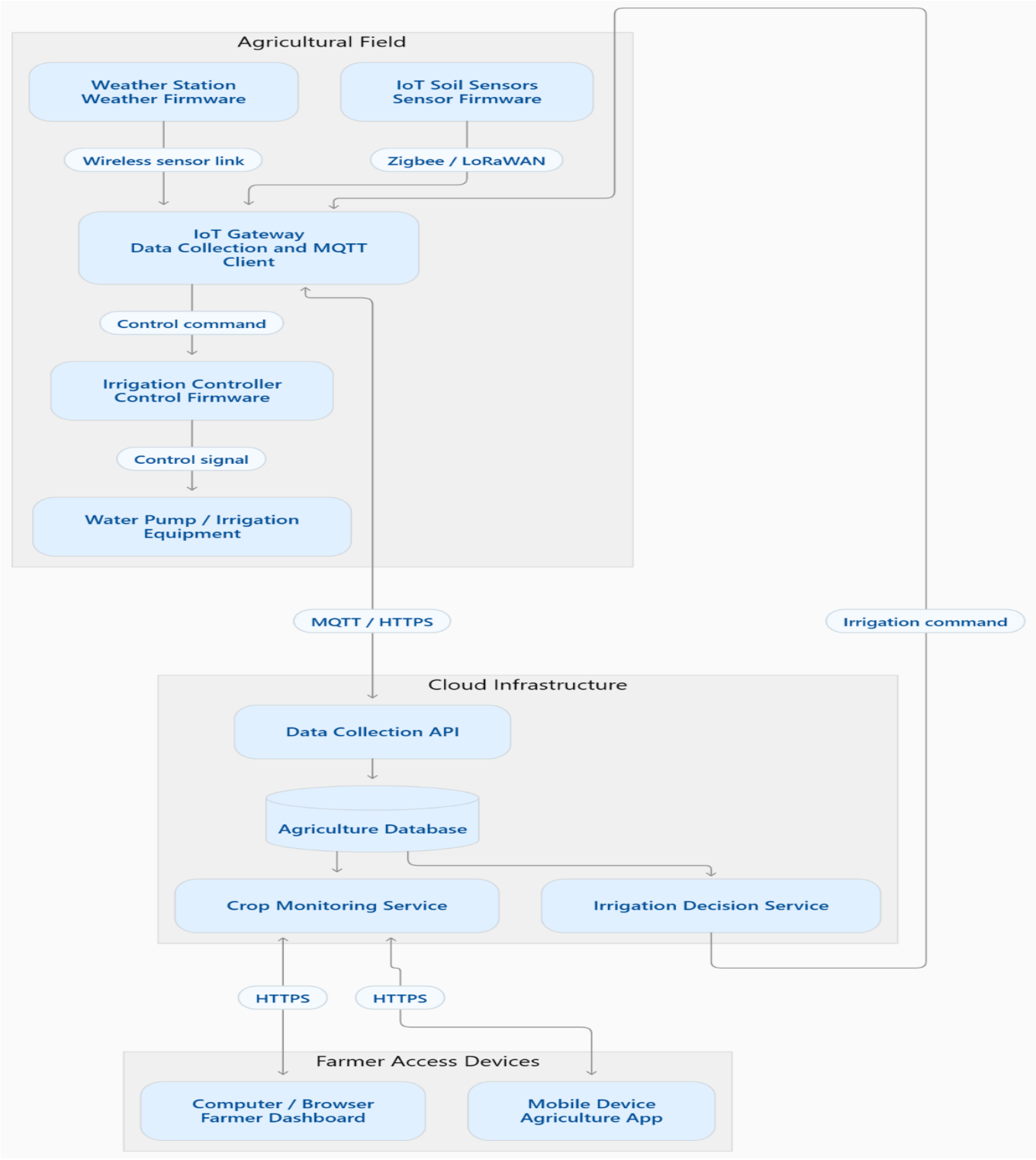
Mobile Device	Mobile agriculture application
Farmer Dashboard	Web dashboard application and browser

Communication protocols

- Sensors to gateway: Zigbee, LoRaWAN, or another supported sensor-network protocol.
- Gateway to cloud: MQTT over a network connection or HTTPS.
- Mobile device and dashboard to cloud: HTTPS.
- Cloud to irrigation controller: MQTT or another supported control protocol through the gateway.

The exact protocols depend on the deployed hardware and network configuration.

Deployment Diagram



RUBRICS FOR CALCULATION

Application Based Questions

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Understanding of Problem Context	20	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand